

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 17-03-26
Time: 3 Hrs Max. Marks: 300

17-03-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-8_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

Rankers Academy JEE
THE NARAYANA GROUP

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PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A transformer operating at primary voltage 8 kV and secondary voltage 160 V serves a load of 80 kW . Assuming the transformer to be ideal with purely resistive load and working on unity power factor, the loads in the primary and secondary circuit would be

   A) 800Ω and 1.06Ω B) 10Ω and 500Ω

   C) 800Ω and 0.32Ω D) 1.06Ω and 500Ω

2. In the experiment for measurement of viscosity η of given liquid with a small ball having radius R, consider following statements.

   (1) Viscosity η depends upon initial velocity of projection of small ball

   (2) Increase in temperature of liquids increases its viscosity

   (3) Increase in temperature of gases increases its viscosity

   (4) Two aluminium spheres of radii R and 2R dropped in a liquid, then their terminal velocities will be in the ratio 1 : 4

   The correct statements are

   A) 2 & 3 only B) 1, 2 & 3 only C) 3 & 4 only D) 1, 2 & 4 only

3. Consider following statements

   (1) Surface tension of bubble is more than that of drop of same radius and material

   (2) Mean free path of a gas molecule will be quadrupled if its number density is halved and size is doubled

   (3) Bernoulli's equation can be applied for all types of fluids

   (4) Two streamlines can never intersect

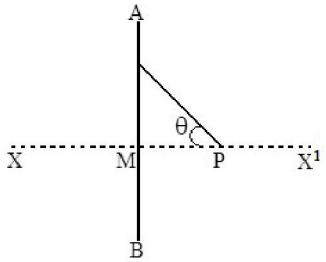
   (5) Angle of contact for Hg & glass is obtuse.

   The correct statements are

   A) 3 & 4 only B) 1 & 4 only C) 2, 3, 4 & 5 only D) 4 & 5 only

4. A wire AB of length l with linear charge density $\lambda = \lambda_0\theta$ as shown.

   (θ taken from $X - X^1$) $\angle MPA = \alpha, \angle MPB = \beta, MP = r$



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Correct value of electric field at point P in the direction x-x^1 is $\left(k=\frac{1}{4\pi\varepsilon_0}\right)$

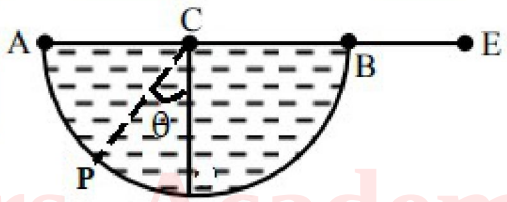
A) $\frac{k\lambda_0}{r}[\alpha\sin\alpha-\cos\alpha-\beta\sin\beta-\cos\beta)]$

B) $\frac{k\lambda_0}{r}[(\alpha+1)\sin\alpha-(\beta+1)\sin\beta)]$

C) $\frac{k\lambda_0}{r}[\alpha\sin\alpha+\cos\alpha-\beta\sin\beta-\cos\beta)]$

D) $\frac{k\lambda_0}{r}[\alpha\sin\alpha-\cos\alpha-\beta\sin\beta+\cos\beta)]$

5. A hemispherical vessel is completely filled with liquid of refractive index μ. A small coin is fixed at P as shown in figure. The min value of refractive index of liquid so that person can see the coin from point E (at the level of vessel) is $(\theta=30^0)$



A) $\sqrt{3}$ B) 3/2 C) $\frac{2}{\sqrt{3}}$ D) $\sqrt{3}/3$

6. A long straight wire carrying current i, a particle having positive charge q and mass m kept at a distance x_0 from wire is projected away from it with speed v normal to it. Maximum separation between wire and particle is _____

A) $x=x_0e^{\left(\frac{2\pi mv}{\mu_0qi}\right)}$ B) $x=x_0e^{\left(\frac{-4\pi mv}{\mu_0qi}\right)}$ C) $x=x_0\ln\left(\frac{2\pi mv}{\mu_0qi}\right)$ D) $x=x_0\left(1-\frac{2\pi mv}{\mu_0qi}\right)$

7. A reversible engine operates between temperatures 227^{0}C and T. The energy rejected from this engine is received by second reversible engine at the same temperature T. the second engine rejects energy at temperature 127^{0}C. Find temperature T as the engines produce the same amount of work output is

A) 277^{0}C B) 177^{0}C C) 200^{0}C D) 150^{0}C

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8. **Assertion (A):** Elasticity of steel is more than rubber.

Reason (R): The speed of sound in a medium is inversely proportional to the square root of its density $\left(v\alpha \frac{1}{\sqrt{\rho}} \right)$.

A) Both A and R are true and R is the correct explanation of A.

B) Both A and R are true but R is NOT the correct explanation of A.

C) A is true but R is false.

D) A is false but R is true.

9. Choose the incorrect statement:

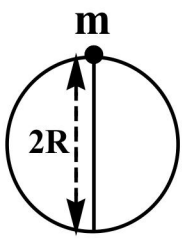
A) The image of sun can never be produced at the center of a glass sphere on earth, (assuming paraxial approximation and the space between the sphere and sun is filled with air)

B) If a lens having $\mu = 1.5$ has focal length 20 cm in air, then its focal length will be 60 cm when it is immersed in water of refractive index $\frac{4}{3}$

C) The laws of reflection and refraction are true for all surface and pairs of media at the point of incidence.

D) The thick lenses give coloured images due to dispersion of light.

10. A bead of mass m slides without friction on the wall of a vertical circular hoop of radius R as shown in figure. The bead moves under the combined action of gravity and massless elastic string attached to the bottom of the hoop. The natural length of the string is R. If the bead is released from top of the hoop with (negligible) zero initial speed, normal reaction on bead, when the bead reaches at bottom point, would be (elastic constant of string k, g is acceleration due to gravity)



A) 5mg B) 5mg + kR C) $\frac{5mg}{2} + kR$ D) $\frac{7mg}{2} + kR$

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11. Given below are two statements: one is labeled as

Assertion A and the other is labeled as Reason R

Assertion A: In a central force field, the work done is independent of the path chosen

Reason R : Potential energy is defined for central forces only.

In the light of the above statements, choose the most appropriate answer from the options given below

A) A is true but R is false

B) Both A and R are true but R is not the correct explanation of A

C) Both A and R are true and R is the correct explanation of A

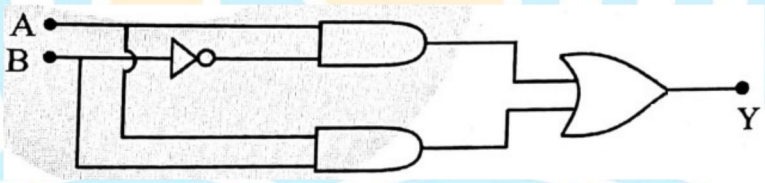
D) A is false but R is true

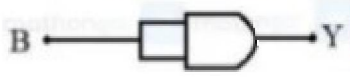
12. Choose the correct nuclear process from the below options

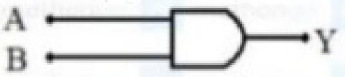
[p: proton, n : neutron, e^- : electron e^+ : positron, υ : neutrino, $\bar{\upsilon}$: antineutrino]

A) $p \rightarrow n + e^- + \bar{\upsilon}$ B) $p \rightarrow n + e^- + \upsilon$ C) $p \rightarrow n + e^+ + \bar{\upsilon}$ D) $p \rightarrow n + e^+ + \upsilon$

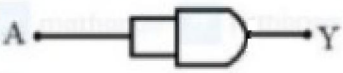
13. Which of the following circuits has the same output as that of the given circuit ?



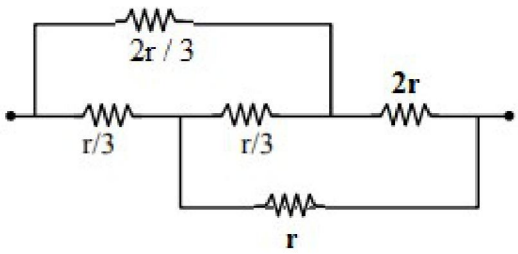
A) 

B) 

C) 

D) 

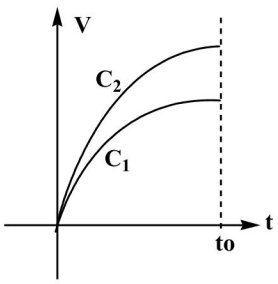
14. Find the equivalent resistance between two ends of the following circuit.



A) 4r B) r / 6 C) 8r / 9 D) 3r / 9

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15. Two capacitors C_1 and C_2 are connected series to a battery. Voltage-time graph is shown below for the two capacitors. The energy stored with them are U_1 and U_2 respectively. Which of the given statements is true ?



A) $C_1 > C_2, U_1 > U_2$ B) $C_2 > C_1, U_2 < U_1$

C) $C_1 > C_2, U_1 < U_2$ D) $C_2 > C_1, U_2 > U_1$

16. Due to presence of an EM-wave whose electric component is given by $E = 100 \sin(\omega t - kx) NC^{-1}$, a cylinder of length 1 cm and radius 1 cm holds certain amount of EM-energy inside it. If another sphere kept in the same EM Wave holds $\frac{9}{16}$ of EM energy of that of the cylinder mentioned above then find radius of the sphere (given that wavelength is very less with respect to size of the cylinder and sphere)

A) 0.75 cm B) 0.25 cm C) 0.5 cm D) 1 cm

17. A short electric dipole of dipole moment $\vec{P}$ and moment of inertia I, kept at origin on a smooth horizontal surface. The dipole moment of dipole makes an angle 60 with x-axisand lies in horizontal plane. If an external uniform electric field E is switched on along x-axis the angular velocity acquired by the dipole, when it becomes parallel to x-axis is

A) $\sqrt{\frac{2PE}{I}}$ B) $\sqrt{\frac{PE}{I}}$ C) $\sqrt{\frac{PE}{2I}}$ D) $\sqrt{\frac{4PE}{2I}}$

18. A proton of mass m_p has same debroglie wave length as that of photon of energy E. The kinetic energy of proton is

A) E B) $\frac{E^2}{m_p c^2}$ C) $\frac{2E^2}{m_p c^2}$ D) $\frac{E^2}{2m_p c^2}$

19. The linear density of a rod of length l and uniform cross section varies with the distance x from left end as $\lambda = \lambda_0 x^2$. The distance of centre of mass of rod from left end is

A) $l/3$ B) $2l/3$ C) $3l/4$ D) $5l/6$

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20. The power of combination of two thin lenses in contact is + 1.0 D. If the focal length of one of the lenses of the combinations +20.0 cm, the focal length of the other lens would be:

A) -120.0 cm B) +80.0 cm C) -25.0 cm D) -20.0 cm

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A tiny metallic cylindrical structure has length and radius of 5 mm and 2.5mm, respectively. Using a specially designed screw gauge which has pitch of 0.75 mm and 15 divisions in the circular scale. You are asked to find the volume of the cylinder. In this measurement, the maximum fractional error will be $\frac{x}{100}$ where x is _____

22. Three rigid bodies, a thin circular ring, a uniform solid sphere and a uniform disc each of mass M, are made to rotate about a common tangent (parallel to their diameter and lying in their planes. The ring and disc have same radius R. The sphere has radius KR, Now, I_{disc}, I_{ring}, I_{sphere} represents moment of inertia of disc, ring and sphere about that common tangent respectively.

If, $I_{sphere} = 35/22\ (I_{disc} + I_{ring})$, find value of $\sqrt{8K^2}+1$

23. In a measurement, it is asked to find coefficient of viscosity per unit force applied on the system. The measured quantity has dimension of $\left[M^aL^bT^c\right]$.

The value of $|a|+|b|+|c|$ is

24. A string of area of cross-section $4\,mm^2$ and length $0.5\,m$ is connected with a rigid body of mass 2 kg . The body is rotated in a vertical circular path of radius $0.5\,m$. The body acquires a speed of $5\,m/s$ at the bottom of the circular path. Strain produced in the string when the body is at the bottom of the circle is ____ $\times 10^{-5}$. (Use Young's modulus $10^{11}\,N/m^2$ and $g = 10\,m/s^2$)

25. A thin film of a specific material can be used to decrease the intensity of reflected light. There is destructive interference of waves reflected from upper and lower surfaces of the film. These films are called anti- reflection coating. Magnesium fluoride (MgF) is used as anti-reflecting coating on plane glass surface for light having wavelength 500 nm and normal incidence. The minimum thickness of film required is 100 nm. Find the value of 100 times of refractive index of magnesium fluoride. It is known that refractive index of glass is more than refractive index of magnesium fluoride.

For More Join: @IITJEE_Advanced

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Statement-I: CFSE { Crystal field stabilising Energy} of $[MnF_6]^{-4}$ is greater than $[CrF_6]^{-3}$

Statement-II: In octahedral crystal field, the t_{2g} orbitals are raised by $0.6\Delta_0$

In light of above statements, choose the correct answers from the following options:-

A) Statement I is true but statement-II is false

B) Statement I is false but statement-II is true

C) Both statement I & II are true

D) Both statement I & II are false

27. The bond length order of following

$\underset{(A)}{C-O}$ $\underset{(B)}{C\equiv O}$ $\underset{(C)}{N-O}$ $\underset{(D)}{N=O}$

A) $A>B>C>D$ B) $A>B>D>C$

C) $A>C>D>B$ D) $C>A>D>B$

28. Among the following compound, which do (es) not exhibits both optical as well as geometrical isomerism

P) 3-chloro hex-3-ene Q) 2-methyl pent-3-en-2-ol

R) 2-methyl pent-2-en-3-ol S) 3-methyl pent-3-en-2-ol

A) Only R B) P and S only C) P and R only D) P, Q and R only

29. Identify the correct statements

(I) Sodium dichromate is more soluble than potassium dichromate

(II) sodium dichromate is used as primary standard in volumetric analysis

(III) Sodium & Potassium chromates are strong reducing agents

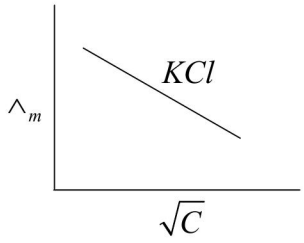
(IV) Chromate & Dichromate are inter convertible in aqueous medium depending on pH of solution

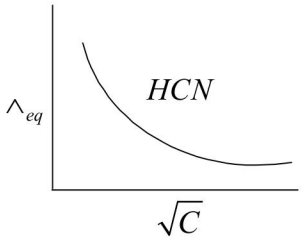
A) I,II,III B) I only C) I & II D) I & IV

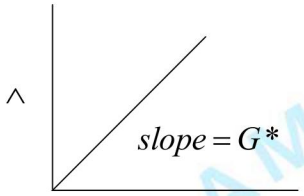
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30. Which of the following graph is correct?

$\wedge$ = conductivity $\wedge_m$ = molar conductivity $\wedge_{eq}$ = equivalent conductivity C = concentration, G = conductance, G* = Cell constant

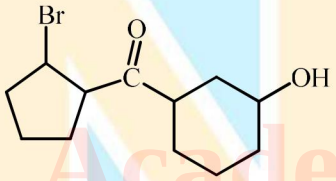
A) 

B) 

C) 

D) All of these

31. Statement I – Compound (A) give below, does not dissolves in $NaHCO_3$ solution & is optically active



Statement II: Compound (B) given below, contains 5 atoms in plane & 3 atoms out of the plane $H_2C=C=C=CH_2$

In light of above statements, choose the correct answers from the following options:-

A) Statement I is true but statement-II is false

B) Statement I is false but statement-II is true

C) Both statement I & II are true

D) Both statement I & II are false

32. **Statement I:** The order of dipole moment is $CH_3F > CH_3Cl > CH_3Br > CH_3I$

Statement II: PbI_4 does not exists because $Pb-I$ bond initially formed during reaction does not release enough energy to unpair $6s^2$ electrons and excite one of them to higher orbital to have 4 unpaired electrons around lead atom.

A) Statement I is true but Statement-II is false

B) Statement I is false but Statement-II is true

C) Both statements I & II are true

D) Both statements I & II are false

For More Join: @IITJEE_Advanced

33. Lead sulphate is soluble in ammonium acetate solution due to formation of compound of Pb ion. The formula & oxidation state of central metal ion is

A) $(NH)_4\left[Pb(CH_3COO)_6\right], +4$ B) $(NH_4)_2\left[Pb(CH_3COO)_4\right], +2$

C) $(NH_4)_2\left[Pb(CH_3COO)_4\right], +4$ D) $(NH_4)_2\left[Pb(CH_3COO)_6\right], +2$

24. Consider the following spectral lines for atomic hydrogen.

(A) Longest wave length line of paschen series

(B) 3rd Longest wave length line of Balmer series

(C) 2nd Longest wave length line of paschen series

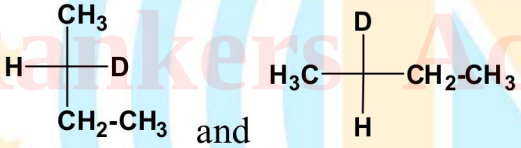
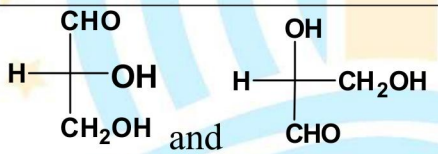
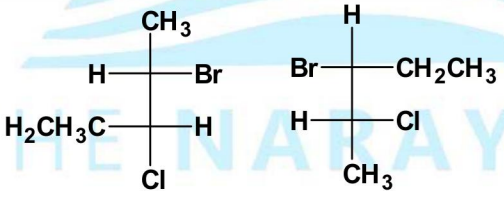
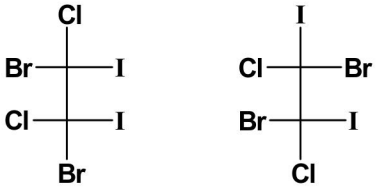
(D) 4th minimum energy line Bracket series

The correct arrangement of above lines in ascending order of energy is

A) $D < A < C < B$ B) $A < C < B < D$

C) $C < D < B < A$ D) $A < B < C < D$

35. Match List-I with List-2

	List-I **Pair of compounds**		**List-2** **Types of isomers**
A)		I)	Identical
B)		II)	Enantiomer
C)		III)	Position isomer
D)		IV)	diasteriomers

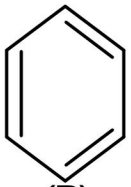
Choose the correct answer from the options given below:

A) A-I, B-II, C-III,D-IV B)A-II, B-I, C-III,D-IV

C) A-I, B-III, C-II,D-IV D) A-III, B-IV, C-I,D-II

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36.



(R)

(i) oleum
(ii) NaOH, Δ
(iii) $Br_2, CS_2, 273\ K$ → major product (P)

Consider the above sequence of reactions. The maximum number of atom(s) possible in a plane in the final product (P electrically neutral) will be:

A) 13 B) 9 C) 12 D) 10

37. Aqueous HCl reacts with $MnO_2(s)$ to form $MnCl_2(aq)$, $Cl_2(g)$ and $H_2O(l)$. What is the weight (in g) of Cl_2 liberated when 17.4g of $MnO_2(s)$ is reacted with 1 lit., of 0.5M aqueous HCl solution? (Given Molar mass in $g\,mol^{-1}$ $Mn = 55$, $Cl = 35.5, O = 16, H = 1$)

A) 8.875 B) 14.2 C) 21.3 D) 71

38. 100ml of mixture of C_2H_4 and C_3H_8 on complete combustion produces 225 ml of CO_2 gas. Volume of oxygen consumed in the process is:

A) 350 ml B) 300 ml C) 250 ml D) 200 ml

39. Calculate the $\Delta_f H^0(in\ Kj)$ of $H_2C = \underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}} - CH = CH_2\ (isoprene)$

Given :-

Energy of C–H bond $= \alpha\ Kj$

Energy of C–C bond $= \beta\ Kj$

Energy of C=C bond $= \gamma\ Kj$

$\Delta^0 H$ atomisation of $C_{(s)} = \delta\ Kj$

$\Delta^0 H_{diss}$ of $H_2 = w\ Kj$

Given Reaction:-

$5C_{(s)} + 4H_2(g) \longrightarrow H_2C{=}\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}{-}CH{=}CH_2$

A) $-5\delta + 4w - 8\alpha - 2\beta - 2\gamma$ B) $-5\delta + 2w - 8\alpha - 2\beta - 2\gamma$

C) $5\delta + 4w - 8\alpha - 2\beta - 2\gamma$ D) $5\delta + 4w - 6\alpha + 2\beta + 2\gamma$

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40. Consider the following reactions

Reaction I : Trans – 2 – pentene $\xrightarrow{Br_2/CCl_4}$ products

Reaction II : cis– 2 – pentene $\xrightarrow{Br_2/CCl_4}$ products

Consider the following statements about reaction I and II

(i) In Reaction I meso products are formed

(ii) In Reaction II racemic mixture is formed

(iii) In Reaction I optically active mixture is produced

(iv) In Reaction II optically inactive mixture is produced

(v) In Reaction I racemic mixture is produced

(vi) Products formed in Reaction I and II are diastereomers to each other

Which of the above statements are incorrect

A) i, ii and iii B) iv and v C) ii and vi D) i and iii

41. NO_2 required for a reaction is produced by the decomposition of N_2O_5 in CCl_4 as per equation,

$2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g).$

The initial concentration of N_2O_5 is $3.00\,mol\,L^{-1}$ and becomes 2.75 mol L^{-1} after 30 minutes. The rate of formation of NO_2 is

A) $1.667\times10^{-2}\,mol\,L^{-1}\,\text{min}^{-1}$ B) $4.167\times10^{-3}\,mol\,L^{-1}\,\text{min}^{-1}$

C) $8.333\times10^{-3}\,mol\,L^{-1}\,\text{min}^{-1}$ D) $2.083\times10^{-3}\,mol\,L^{-1}\,\text{min}^{-1}$

42. Consider the following process/reaction given with the property

(i) Benzoic acid > 2-oxo cyclo hexane carboxylic acid (ease of decarboxylation)

(ii) p-nitro phenol < o-nitro phenol (acidic strength)

(iii) β – diketo compound > β – keto ester compound (percentage of enol content)

(iv) Polar protic solvent > polar aprotic solvent (S_N1 reaction)

Identify the correct properties

A) i and iii B) iii and iv C) ii and iv D) i, iii and iv

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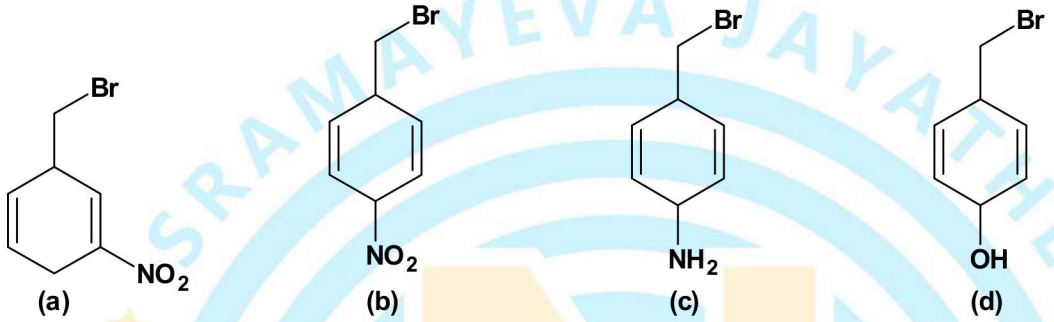
43. The correct order of the rate of the reaction for the following reaction with respect to leaving group is (where: $Lg =$ leaving group)

$$CH_3 - Lg + OH^- \longrightarrow CH_3 - OH + Lg^-$$

A) $RO^- > OH^- > PhO^- > CH_3COO^-$ B) $OH^- > OR^- > PhO^- > CH_3COO^-$

C) $RO^- > OH^- > CH_3COO^- > PhO^-$ D) $CH_3COO^- > PhO^- > OH^- > RO^-$

44. The correct order of reactivity of the following Halides towards reaction with water



A) $d > c > a > b$ B) $d > c > b > a$ C) $c > d > a > b$ D) $c > d > b > a$

45. Given below are two statements:

Statement-I: The correct order of size $K > Na > Mg^{+2} > Li^+$

Statement-II: The correct order in terms of the magnitude of electron gain enthalpy is

$Cl > Br > S > O$

In the light of the above statements choose the correct answer from the option given below

A) Both statement I and statement II are false

B) statement I is false But statement II is True.

C) statement I is true But statement II is False

D) Both statement I and statement II is True

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. First & Second ionisation constants of H_2X are 1.5×10^{-10} & 1.8×10^{-16} respectively. The concentration of X^{-2} in $0.1M$ H_2X solution is _____ $\times10^{-16}M$ {nearest integer}

47. In case of $\left[Cr(CN)_6\right]^{3-}$ complex,

a = number of 90^0 angles around central metal atom

b = maximum number of atoms in a plane

c = number of σ bonds.

Find the value of a + b + c = _____

48. A decinormal solution of $NaCl$ exerts an osmotic pressure of 4.6 atm at 300 K. Calculate its percentage of dissociation $\left[R=0.082\,atm\,K^{-1}mol^{-1}\right]$

49. How may isomers are possible for a compound with formula $\left[Rh(en)_2Cl(NO_2)\right]$

50. MX is sparingly soluble salt that follows the given solubility equilibrium at 298 K

$MX(s)\rightleftharpoons M^+(aq)+X^-(aq)\;K_{sp}=10^{-16}$

If standard reduction potential for $M^+(aq)\xrightarrow{+e^-}M(s)$ is 0.49 then value (magnitude) of standard reduction potential for the metal/metal insoluble salt electrode $E^0_{X^-/MX(s)/M}$ is _____ mV {nearest integer}

[Given that $\frac{2.303RT}{F}=0.059V$]

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Considering only the principal values of the inverse trigonometric functions, the number of values in $[-4\pi, 6\pi]$ for which $2\tan^{-1}(\sin^3 x) = \tan^{-1}(\tan^2 x)$ is

A) 0 B) 22 C) 16 D) 11

52. If $a,b,c \in \mathbb{N}$ (i.e the set of natural numbers), and Let $R_1 = \{(x,y) : y > ax^2 + bx + c\}$,

$R_2 = \{(x,y) : y = ax^2 + bx + c\}$
$R_3 = \{(x,y) : y < ax^2 + bx + c\}$,

Then which of the following options is/are INCORRECT?

A) All points in Quadrant-IV lie in R_3

B)If all points on the circle $x^2 + y^2 = 1$ lie in R_3 ,minimum possible value of $a+b+c$ is 3

C) $R_3 \cup (2,1) = R_3$

D) $R_1 \cap (0,0) = \varnothing$

53. Let $P(2,2,4)$ be a point in space and $L : \frac{x-1}{2} = \frac{y+1}{1} = \frac{z-2}{2}$ be a line. Let Q_1 and Q_2 be two distinct points on the line L such that the distance PQ_1 and PQ_2 are both equal to $\sqrt{14}$. The area of the triangle PQ_1Q_2 is equal to:

A) $3\sqrt{5}$ B) $6\sqrt{5}$ C) $4\sqrt{6}$ D) $5\sqrt{3}$

54. Number of one-one functions from $\{1,2,3,4,5\} \to \{1,2,3,4,5,6,7\}$ such that $f(i) \le f(i+2) \forall\ i = 1,2,3$ is

A) 21 B) 105 C) 250 D) 210

55. Let z be a complex number satisfying $\frac{\pi}{4} \le \arg\left(\frac{z-1}{z-i}\right) < \pi$ and $|z-2-i| \le \sqrt{2}$, then sum of least and greatest value of $|z-i|$ is

A) 2 B) $4\sqrt{2}$ C) $4-\sqrt{2}$ D) $4+\sqrt{2}$.

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56. A ray of light emerging from the point $A(1,2)$ is reflected at a point P on the x-axis and then passes through the point $B(5,3)$. The coordinates of the point of reflection P are:

A) $\left(\frac{13}{5},0\right)$ B) $\left(\frac{11}{5},0\right)$ C) $\left(\frac{14}{5},0\right)$ D) $\left(\frac{12}{5},0\right)$

57. $f(x)=\ln(1+|x|)-\frac{|x|}{1+x};x\in\mathbb{R}-\{-1\}$, then the. number of points in domain at which. $f(x)$ attains local extrema is k_1 and. number of stationary points is k_2 then $k_1+k_2=$

A) 2 B) 3 C) 1 D) 4

58. The area (in sq.units) of the region $R=\{(x,y):x^2\le y\le x \text{ and } [x+y]=0\}$, where [.] denotes the greatest integer function, is:

A) $\frac{10\sqrt{5}-23}{24}$ B) $\frac{5\sqrt{5}-10}{12}$ C) $\frac{10\sqrt{5}-17}{24}$ D) $\frac{5\sqrt{5}-13}{12}$

59. A bag contains 10 balls numbered 1 to 10 each. A person takes out three balls one by one with replacement. If there exists atleast one even number among the three balls he adds all the 3 numbers and reports it. If there exists no even number among the three balls he multiplies all the 3 numbers and reports it. If he reports 9, then the probability that all the three balls have odd numbers on them is

A) $\frac{1}{3}$ B) $\frac{1}{4}$ C) $\frac{1}{2}$ D) $\frac{3}{4}$

60. If $(x-3)^2+(y-1)^2=r^2$ has 3 common points with the curve $yx-x^2=0$, then the sum of all distinct values of $|r|$ is

A) $\sqrt{2}\left(\sqrt{5}+1\right)$ B) $\sqrt{10}+3$ C) $3\sqrt{2}$ D) 1

61. Let PQ be a focal chord of the parabola $y^2=8x$. If the area of triangle OPQ (where O is the vertex of parabola) is $12\sqrt{2}$sq. units, then the length of the focal chord PQ is

A) 24 B) 32 C)36 D) 48

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62. If the curve $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ where $a>b>0$ has eccentricity 'e' and is confocal with the curve $\frac{2x^2}{a^2}-\frac{2y^2}{b^2}=1$, then 'e' is

A) $\frac{1}{2}$ B) $\sqrt{3}$ C) $\sqrt{\frac{2}{3}}$ D) $\frac{1}{\sqrt{3}}$

63. The number of distinct real values of x in $[-5\pi,5\pi]$ which satisfy the equation $\sin x-2\cos 2x+\sin 3x=2$, is

A) 5 B) 10 C) 11 D) 6

64. $\int_0^{\frac{\pi}{4}}\frac{\tan x+(\tan x)^2}{1+(\tan x)^2}dx=$

A) $\frac{\pi}{8}$ B) $\frac{\pi}{4}$ c) $\frac{\pi}{4}+\ln 2$ D) $\frac{\pi}{8}+\ln 4$

65. Let a twice differentiable function $y=f(x)$ defined from $R\to R$ satisfy $(x^2+1)\frac{dy}{dx}=-2x(\tan^{-1}y)(1+y^2)$ and $\lim_{x\to 0}\frac{f(x)-(\tan 1)}{x^2}=L$ which is a finite non-zero real number, then $\left|L\left(\tan^{-1}(f(1))\right)\right|=$

A) $\frac{\tan^2 1}{2}$ B) $2\sec^2 1$ C) $\frac{\sec^2 1}{2}$ D) $\sec^2 1$

66. If $P=\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$; $Q=\begin{bmatrix}3&0&0\\0&-3&0\\0&0&2\end{bmatrix}$; $M=PQP$, then the sum of principal diagonal elements of the matrix M^9, is

A) 512 B) 1024 C) $2(3^9)+2^9$ D) 0

67. Let $2x-3y+z=4$, $x-3y+2z=3$, $x+3y+q=pz$ has infinite solutions for (x,y,z), then the remainder obtained when $(p+q)^{50}$ is divided by p^2, is

A) 9 B) 5 C) 17 D) 12

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68. A data with it's frequencies is given as.

x_i :	1	2	3
$f(x_i)$:	f_1	f_2	f_3

such that $|f_r - r| \le 1 \ \forall r = 1,2,3$ Then minimum mean of the data is

A) 1 B) 2 C) 3 D) $\frac{3}{2}$

69. A relation ' R ' defined on $\{1,2,3,4,5\}$ is such that $(x,y) \in R$ if $2y < x^2 + 1$. Minimum number of elements to be added to make it reflexive and transitive is

A) 1 B) 2 C) 10 D) 9

70. Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors such that $|\vec{a}| = 2$ and $|\vec{b}| = 3$, and the angle between them is 60^0. Let $\vec{p}$ be a vector satisfying the conditions $\vec{p} \times \vec{b} = \vec{a} \times \vec{b}$ and $\vec{p} \,.\, \vec{a} = 0$. Let $\vec{q}$ be another vector satisfying the conditions $\vec{q} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{q}.\vec{b} = 0$. Then the exact value of the dot product $\vec{p}.\vec{q}$ is

A) -9 B) -12 C) 9 D) 12

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. If $(1+2x)^{20} = a_0 + a_1 x + a_2 x^2 + \ldots + a_{20} x^{20}$. where a_r denotes the coefficient of x^r and $3a_0 + 2a_1 + 3a_2 + 2a_3 + 3a_4 + 2a_5 + \ldots + 2a_{19} + 3a_{20} = \frac{M(3^{20})+1}{2}$, then M is

72. Let ' f ' be a differentiable function in $\mathbb{R}$ such that $f(x+y) = f(x) + f(y) + xy(3x+3y-2) \forall x, y \in R$ and $f'(1) = 1$, then $f'(-2) - f(-2) =$

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73. Let $f(x)=(\sqrt{5}+1)^x-(\sqrt{5}-1)^x; x>0$ and if $\lim\limits_{x\to\infty}\dfrac{f(x+2)}{f(x)}=a$, then $\left[\sqrt{a}\right]=$ _____

($[.]$ denotes G.I.F)

74. Let ' p ' be an integer for which both roots of the quadratic equation $x^2+2(p-3)x+9=0$ lies in $(-6,1)$. If $20,g_1,g_2,g_3,\ldots,g_{19},g_{20},p$ are in G.P, then $\sum\limits_{r=1}^{20}g_r g_{21-r}=$_____

75. Number of integral values of x satisfying the inequality $\log_{\left(x+\frac{5}{2}\right)}\left(\dfrac{x-5}{2x-3}\right)^2>0$ is___

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